

An Advanced Retrieval-Augmented Generation Framework with Chart Insight Extraction for the Vietnamese Financial Market

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Abstract—Retrieval-Augmented Generation (RAG) has become an effective approach for reducing hallucinations in Large Language Models (LLMs) by grounding responses in external knowledge. However, existing financial RAG systems primarily rely on Optical Character Recognition (OCR), which extracts textual content while discarding the semantic information embedded in financial charts. Consequently, chart-oriented queries often suffer from incomplete retrieval contexts and inaccurate responses.

This paper presents a chart-aware RAG framework for Vietnamese financial reports and systematically investigates three retrieval strategies, ranging from conventional OCR-based retrieval to multimodal retrieval and a proposed semantic indexing approach that converts charts into structured textual representations using chart understanding models. The extracted chart semantics are indexed jointly with OCR text, enriching the retrieval corpus without increasing inference complexity.

Experiments on a benchmark of approximately 200 expert-designed financial question-answer pairs demonstrate that the proposed approach achieves the best overall performance, improving the average Fact Fidelity from 0.320 to 0.791 and the average Insight Relevance from 0.293 to 0.712 compared with the strongest baseline. Furthermore, the proposed method reduces score variance by over 85%, indicating substantially more stable retrieval performance. Statistical analysis using the Wilcoxon signed-rank test confirms that the improvements are highly significant ($p < 10^{-6}$). These results demonstrate that incorporating chart semantics during indexing is considerably more effective than relying solely on multimodal reasoning during inference for financial document understanding.

Index Terms—Retrieval-Augmented Generation (RAG), Financial Document Intelligence, Chart Understanding, Dynamic Chart Synthesis.

I. INTRODUCTION

Large Language Models (LLMs) have improved question answering across many domains, and Retrieval-Augmented Generation (RAG) has become a standard approach for reducing hallucination by grounding responses in external documents. In RAG, the model retrieves relevant evidence from a knowledge base and uses it during generation, which improves factual consistency and reliability.

Financial question answering is a natural fit for RAG because investment analysis depends on up-to-date and evidence-based information. Analysts rely on annual reports, quarterly

reports, earnings presentations, and regulatory filings to evaluate company performance and market trends. As a result, many financial chatbot systems now use RAG to support investment-related queries.

However, most existing Financial RAG systems are designed for text-only documents and do not fully exploit the multimodal structure of financial reports. These reports usually contain narrative text, tables, and charts. In many cases, charts carry the most important information because they summarize revenue trends, profit growth, cash flow changes, and comparisons across periods. Yet this visual information is often ignored during indexing.

The common OCR-based pipeline extracts text from report pages using OCR engines such as EasyOCR or PaddleOCR, splits the text into chunks, embeds them, and stores them in a vector database. While efficient, this approach treats financial reports as plain text. Charts are reduced to axis labels or legends, and their numerical patterns and visual trends are lost. As a result, questions that require chart interpretation are difficult to answer correctly.

A simple alternative is to keep chart images during retrieval. Recent multimodal RAG systems detect charts, store cropped images together with text embeddings, and send both text and images to a Vision-Language Model (VLM) at inference time. This preserves visual content, but it also has two limitations. First, chart understanding is delayed until inference, so the same chart must be interpreted repeatedly for similar queries. Second, VLMs reason over pixels rather than structured data, which increases computation and can weaken numerical accuracy.

These limitations suggest that the main bottleneck is not only retrieval, but also document representation before indexing. OCR-based systems lose chart semantics, while image-based systems keep charts only as raw visual objects. Neither approach converts charts into explicit knowledge that can be searched effectively.

To address this problem, we propose a chart-aware knowledge representation framework for Financial RAG. Financial reports are first parsed into textual and visual regions using layout analysis. Text regions are processed by OCR, while chart regions are detected by an object detection model. Each chart is then interpreted by chart understanding models such

as Matcha or DePlot, or by VLMs, to extract chart type, title, numerical series, summaries, and key insights. The extracted chart knowledge is fused with OCR text to build enriched document representations that are embedded and stored in the retrieval database. In this way, charts become searchable knowledge units rather than passive images.

To study the effect of document representation, we design three Financial RAG paradigms. The first is a conventional OCR-based text retrieval pipeline that discards chart information. The second preserves chart images and performs multi-modal retrieval over text and visuals. The third is our proposed framework, which converts charts into structured semantic knowledge before indexing and enables unified retrieval over enriched text representations. This setup allows us to compare how different levels of document understanding affect retrieval and reasoning performance.

We evaluate the proposed approach on a Vietnamese financial benchmark built from annual and quarterly reports of publicly listed companies across multiple sectors. The benchmark includes textual documents, chart annotations, and question-answer pairs covering numerical extraction, chart interpretation, cross-document reasoning, and investment analysis. Experiments show that richer document representation improves retrieval accuracy, factual consistency, and financial reasoning while remaining practical for deployment.

The implementation, benchmark, and evaluation scripts are publicly available at:

<https://github.com/tuantung26/VN-FinRAG>

The main contributions of this paper are summarized as follows:

- 1) We formulate three progressively richer knowledge representation paradigms for Financial Retrieval-Augmented Generation, enabling a systematic study of how document representation affects retrieval and reasoning.
- 2) We propose a chart-aware document representation framework that converts financial charts into structured semantic knowledge by combining OCR, chart detection, and chart understanding models before indexing.
- 3) We introduce a unified knowledge fusion strategy that integrates OCR text and chart-derived knowledge into enriched document representations for dense retrieval without repeated visual reasoning at inference time.
- 4) We construct a Vietnamese financial benchmark consisting of financial reports, chart annotations, and analytical question-answer pairs to support future research on multimodal Financial RAG.
- 5) Experimental results show that the proposed chart-aware representation significantly improves retrieval quality and financial reasoning over conventional OCR-based and image-preserving baselines.

II. RELATED WORK

The proposed framework lies at the intersection of Financial Retrieval-Augmented Generation, Document Intelligence,

and Chart Understanding. This section reviews representative studies in each area and discusses their limitations with respect to financial document representation.

A. Financial Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has become the dominant paradigm for enhancing the factual reliability of Large Language Models by retrieving external knowledge during inference. Instead of relying solely on parametric memory, RAG augments the language model with evidence retrieved from external databases, thereby reducing hallucination and improving response accuracy.

Recent studies have extended RAG to domain-specific applications such as healthcare, law, finance, and scientific literature. In the financial domain, RAG has been employed to build intelligent assistants capable of answering questions over annual reports, earnings reports, and market disclosures. Most existing Financial RAG systems adopt a conventional pipeline consisting of document parsing, OCR, text chunking, dense embedding, and vector retrieval.

Although these approaches achieve satisfactory performance on textual question answering, they generally assume that financial documents consist primarily of text. As a result, charts are either discarded during OCR or treated as auxiliary images that are interpreted only during inference. Consequently, visual information rarely contributes directly to document indexing or dense retrieval.

B. Document Understanding for Financial Reports

Financial reports are complex multimodal documents composed of paragraphs, tables, charts, and hierarchical layouts. Recovering structured information from these documents has been an active research topic within Document AI.

Traditional pipelines employ OCR engines such as EasyOCR and PaddleOCR to recover textual content from document pages. More recent document understanding models, including LayoutLM, Donut, MinerU, and Docling, incorporate layout information to improve structural parsing of complex documents. These methods significantly enhance text extraction quality and document reconstruction.

However, most existing document understanding systems primarily focus on textual recovery. Charts are usually recognized as graphical regions without recovering their underlying numerical semantics. Consequently, although document layouts are preserved, the semantic information encoded within charts remains unavailable to downstream retrieval systems.

C. Chart Understanding

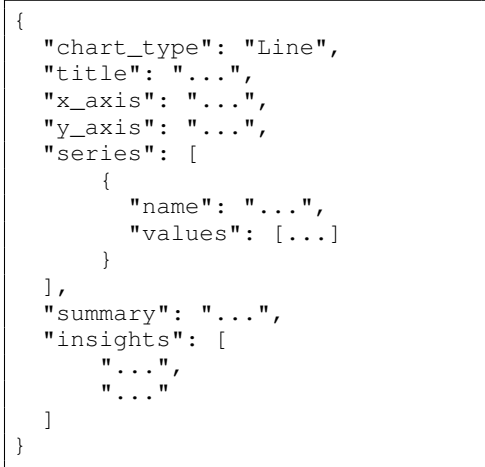


Fig. 1. Example of the structured chart knowledge object generated by the chart understanding module.

Automatic chart understanding has recently attracted considerable attention due to the increasing prevalence of data visualizations in scientific and financial documents. Early approaches relied on OCR and heuristic rules to recover chart labels and axis values. More recently, transformer-based models such as DePlot, Matcha, and UniChart have demonstrated remarkable capability in recovering structured numerical data directly from chart images.

Meanwhile, recent Vision-Language Models (VLMs), including GPT-4o and Qwen2.5-VL, have shown promising zero-shot reasoning ability on chart interpretation tasks without task-specific training.

Despite these advances, existing chart understanding methods are primarily designed for visual question answering or standalone chart analysis. Their outputs are rarely integrated into Retrieval-Augmented Generation systems as searchable semantic knowledge. Consequently, chart understanding is typically performed after retrieval rather than during document indexing.

D. Knowledge Representation for Retrieval-Augmented Generation

Recent research has increasingly recognized that retrieval quality depends not only on retrieval algorithms but also on document representation. GraphRAG, LightRAG, and related methods enrich textual documents by constructing structured knowledge representations prior to retrieval, enabling more effective reasoning over complex relationships.

Inspired by this observation, our work investigates semantic enrichment from a different perspective. Instead of enriching textual entities alone, we focus on transforming charts into structured semantic knowledge before indexing. Unlike previous multimodal RAG systems that retrieve raw chart images, the proposed framework converts chart contents into textual representations containing metadata, numerical values, semantic summaries, and analytical insights. These enriched

representations are jointly embedded with OCR-extracted text, allowing chart semantics to participate directly in dense retrieval.

Table ?? summarizes the differences between representative approaches and the proposed framework.

III. METHODOLOGY

A. Framework Overview

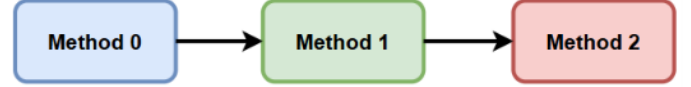


Fig. 2. Method Flow

The objective of this work is not to compare different retrieval algorithms, but to investigate how document representation affects retrieval quality in Financial Retrieval-Augmented Generation (Financial RAG).

The proposed framework consists of three progressively richer document representation paradigms. All three methods share the same embedding model, vector database, retriever, and Large Language Model. Therefore, the only difference lies in how information from financial reports is represented before indexing.

Method 0 performs conventional OCR and discards all chart information.

Method 1 preserves chart images and retrieves them together with textual evidence.

Method 2 further converts charts into structured semantic knowledge before document embedding, allowing chart-derived information to participate directly in dense retrieval.

This progressive design enables a controlled investigation of how semantic representation influences downstream retrieval and reasoning.

B. Method 0: OCR-based Financial RAG

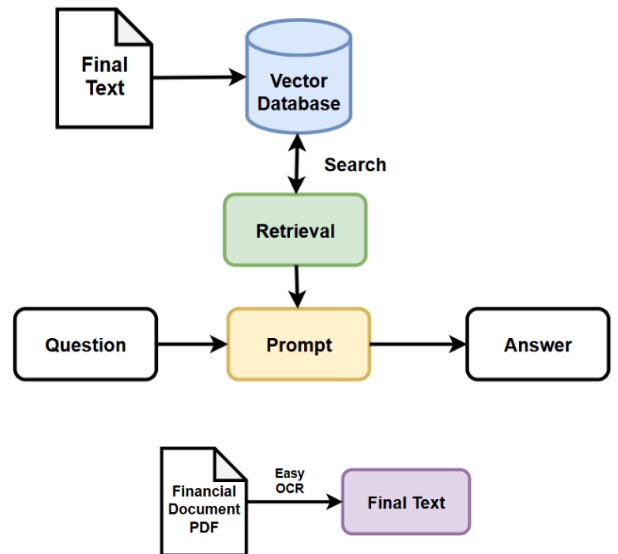


Fig. 3. Pipeline of the conventional OCR-based.

TABLE I
REPRESENTATIVE APPROACHES IN FINANCIAL DOCUMENT UNDERSTANDING

Approach	Text Extraction	Chart Processing	Chart as Knowledge	Pre-index Semantic Enrichment	Cross-modal Retrieval
OCR-based RAG	✓	✗	✗	✗	✗
Multimodal RAG	✓	Image	✗	✗	✓
Chart Understanding	✗	✓	Partial	✗	✗
GraphRAG	✓	✗	✗	Text Only	✓
Our Framework	✓	✓	✓	✓	✓

TABLE II
COMPARISON OF THE THREE PROPOSED FINANCIAL RAG PARADIGMS

Method	OCR Text	Chart Images	Chart Semantics	Embedding Input	Visual Reasoning	Expected Retrieval Quality
Method 0	✓	✗	✗	OCR Text	Not Required	Low
Method 1	✓	✓	At Query Time	OCR Text	Required	Medium
Method 2 (Ours)	✓	✓	Before Indexing	OCR Text + Chart Knowledge	Not Required	High

Given a financial report

$$D = \{P_1, P_2, \dots, P_n\},$$

each page is processed by an OCR engine to extract textual content. In this work, EasyOCR is adopted as the primary OCR engine because of its lightweight deployment and multilingual capability. PaddleOCR is additionally evaluated as an alternative OCR implementation during the experimental analysis. For each page P_i , the OCR engine produces a collection of textual regions

$$T_i = \{t_1, t_2, \dots, t_m\},$$

where each region contains recognized text together with its spatial coordinates. The extracted texts are concatenated, divided into semantic chunks, embedded into dense vectors, and stored inside the vector database. During inference, a user query is embedded into the same vector space, and the retriever selects the Top- k most relevant chunks for answer generation. Although this pipeline is simple and computationally efficient, it assumes that financial reports consist solely of textual information. Charts are reduced to isolated labels or legends, while their numerical relationships, temporal trends, and visual semantics are completely discarded. Consequently, the retriever cannot access chart information, making questions involving trend analysis or numerical comparison difficult to answer correctly. Since chart information is completely absent from the indexed representation, queries referring to numerical trends or visual comparisons cannot be matched through dense retrieval.

C. Method 1: Chart-Preserving Multimodal RAG

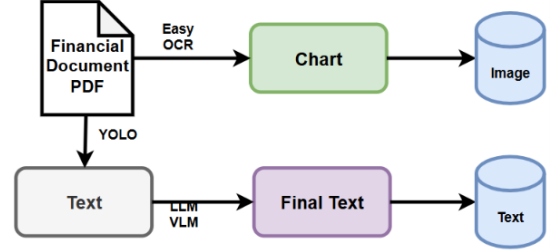


Fig. 4. Pipeline of the chart-preserving multimodal Financial RAG.

To avoid the complete loss of chart information, Method 1 preserves chart images during document indexing. As illustrated, document pages are first analyzed using an object detection model to identify chart regions. In this work, YOLOv8 is employed because of its high detection accuracy and real-time performance on document layouts. After detection, each chart is cropped from the original document page and stored independently as a visual object. Meanwhile, OCR is simultaneously performed on textual regions using the same pipeline described in Method 0. Unlike Method 0, the retrieval database now consists of two complementary repositories:

- textual embeddings extracted from OCR,
- visual chart objects extracted by the detector.

During retrieval, relevant text chunks and chart images are jointly retrieved. The retrieved chart images are subsequently interpreted by a Vision-Language Model (VLM) together with textual evidence to generate the final response. This strategy successfully preserves visual information that would otherwise be discarded. However, chart understanding is postponed until inference time. Whenever similar questions are asked, the same chart must be repeatedly analyzed by the Vision-Language Model, leading to increased computational cost and inference latency. Furthermore, because reasoning

is performed directly on pixels, numerical consistency may vary depending on the capability of the underlying multimodal model. Although chart images are preserved, they remain isolated visual objects stored separately from textual embeddings.

Consequently, the embedding model never observes the semantic content of charts during indexing.

Instead, chart understanding is postponed until inference, where retrieved images are interpreted by a Vision-Language Model together with OCR text.

Therefore, retrieval quality still depends primarily on textual similarity, while chart semantics only become available after retrieval has already finished.

D. Method 2: Semantic-Enriched Chart-aware RAG

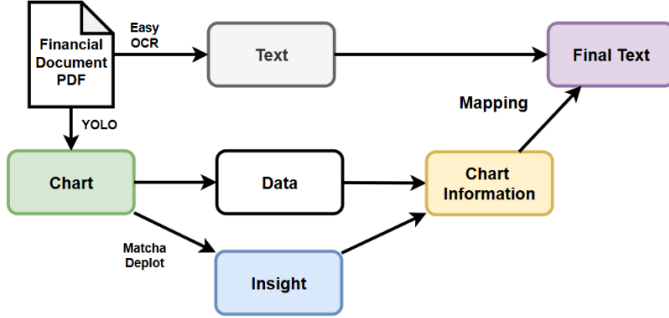


Fig. 5. Pipeline of the proposed semantic-enriched chart-aware Financial RAG.

Method 2 addresses the limitations of both previous approaches by transforming charts into structured semantic knowledge before document indexing, as illustrated. Instead of treating charts as passive images, the proposed framework converts each detected chart into an explicit knowledge object through a dedicated chart understanding module. Following chart detection, each cropped chart is processed using either specialized chart understanding models, such as Matcha or DePlot, or general Vision-Language Models. The objective is not merely to generate captions, but to recover structured information including

- chart title,
- chart type,
- axis descriptions,
- numerical data series,
- semantic summary,
- analytical insights.

Each chart is therefore represented as

$$C = (M, D, S, I),$$

where M denotes chart metadata, D represents numerical data points, S is the semantic summary, and I contains automatically extracted analytical insights. Unlike Method 1, the extracted chart knowledge is converted into textual representations and fused with OCR outputs before document chunking. Consequently, every indexed chunk may simultaneously contain narrative descriptions, numerical values, and chart-derived

semantic knowledge. After knowledge fusion, the enriched documents are embedded using the same embedding model adopted in Method 0 and stored in a unified Milvus vector database. Since chart semantics have already been converted into searchable knowledge, retrieval can be performed entirely within the textual embedding space without invoking additional visual reasoning during inference. Compared with the previous two methods, the proposed representation preserves both textual and visual semantics while maintaining an efficient retrieval pipeline. More importantly, charts become first-class knowledge entities that directly participate in semantic retrieval instead of remaining isolated visual objects. Rather than storing charts as visual objects, the proposed framework transforms every chart into explicit semantic knowledge.

Each chart is represented as

$$C = (M, D, S, I),$$

Unlike raw images, these representations can be directly embedded into the same semantic space as OCR text.

E. Semantic Enrichment Before Indexing

The core hypothesis of this work is that retrieval effectiveness is largely determined by the semantic completeness of document representations.

Dense retrieval relies on vector similarity between the user query and indexed document embeddings.

Consequently, concepts that are absent during embedding cannot be retrieved, regardless of the capability of the downstream language model.

Method 0 embeds only OCR-extracted text.

Method 1 additionally preserves chart images; however, the semantic content of those images is not incorporated into document embeddings because chart understanding is deferred until inference.

In contrast, Method 2 converts charts into structured semantic knowledge before embedding.

Chart titles, numerical values, chart types, semantic summaries, and automatically generated insights are merged with OCR text to construct enriched document chunks.

Formally, let

T

denote OCR text and

C

denote chart knowledge.

The enriched document representation is defined as

$$D_e = T \oplus C,$$

where

$\oplus$

represents the semantic fusion operation.

The embedding model therefore encodes both textual and chart-derived information,

$$E(D_e),$$

instead of

$$E(T).$$

This enriched semantic representation increases contextual completeness, allowing semantically related financial queries to retrieve the correct documents with higher probability.

F. Unified Retrieval

Given a user query

$$q,$$

the embedding model produces

$$E(q).$$

Similarity between the query and enriched document representations is computed using cosine similarity

$$S(q, d) = \cos(E(q), E(D_e)).$$

Top- k documents are retrieved from the Milvus vector database and forwarded to the Large Language Model for answer generation.

G. Chart Semantic Representation

The primary contribution of Method 2 is the transformation of financial charts into structured semantic representations before document indexing. Instead of treating charts as passive visual objects, the proposed framework converts each chart into a textual description that captures both numerical information and high-level analytical insights. Consequently, chart knowledge becomes directly searchable by a conventional text-based retriever.

Given a detected chart image, a chart understanding model (e.g., DePlot) generates a structured representation

$$C = (M, D, I) \quad (1)$$

where

- M denotes chart metadata, including chart title, chart type, axis labels, and measurement units;
- $D = \{d_1, d_2, \dots, d_n\}$ represents the extracted numerical data points;
- I denotes a natural-language summary describing major trends and observations.

Each numerical record is represented as

$$d_i = (\text{label}, \text{value}, \text{series}) \quad (2)$$

where *label* corresponds to the category or timestamp, *value* is the associated numerical quantity, and *series* identifies the corresponding data series.

An example of the generated semantic representation is illustrated in Figure ?? . Unlike raw OCR output, this representation preserves both explicit numerical values and implicit visual trends, allowing financial charts to participate directly in semantic retrieval.

H. Unified Document Index Construction

After OCR extraction, every PDF page consists of textual content obtained by EasyOCR. For Method 2, the semantic descriptions generated from all charts belonging to the same document are appended to the OCR text before indexing.

Formally, the indexed document is represented as

$$D = T \oplus C \quad (3)$$

where T denotes the OCR-extracted text and C denotes the concatenated semantic descriptions generated from all detected charts. The operator $\oplus$ represents document concatenation.

The resulting document is segmented into overlapping chunks using a recursive text splitter,

$$D = \{x_1, x_2, \dots, x_m\}, \quad (4)$$

where each chunk contains approximately 1000 characters with an overlap of 200 characters.

Each chunk is embedded using a dense sentence embedding model

$$e_i = E(x_i), \quad (5)$$

where $E(\cdot)$ denotes the embedding function. The resulting vectors are stored in the vector database for similarity search during retrieval.

Compared with Method 0, the indexed corpus now contains substantially richer semantic information. Queries involving numerical trends, chart comparisons, or visual observations can therefore retrieve more informative contexts even though the retrieval pipeline itself remains unchanged.

I. Retrieval and Response Generation

During inference, a user query is first encoded into the same embedding space as the indexed document chunks.

$$e_q = E(q) \quad (6)$$

Similarity search is then performed over the vector database to retrieve the Top- k most relevant chunks according to cosine similarity,

$$S(q, x_i) = \frac{e_q^\top e_i}{\|e_q\| \|e_i\|}. \quad (7)$$

The retrieved contexts are concatenated and provided to the Large Language Model (Gemini 2.5 Flash) for answer generation.

Unlike Method 1, which relies on the language model to interpret chart images during inference, Method 2 performs chart understanding offline during the indexing stage. Consequently, the retrieved contexts already contain structured chart

semantics, allowing the retriever to match chart-related queries more accurately. This design significantly enriches the retrieval corpus while introducing almost no additional computational cost during inference.

IV. DATASET CONSTRUCTION

A comprehensive benchmark is essential for evaluating Retrieval-Augmented Generation (RAG) systems in the financial domain. Existing public RAG benchmarks mainly focus on general-purpose question answering and contain limited financial documents, while chart-centric benchmarks such as ChartQA and PlotQA are designed for visual question answering rather than document retrieval. Consequently, they are unable to evaluate whether chart semantics can improve document indexing and retrieval in a financial RAG pipeline.

To address this limitation, we construct a domain-specific benchmark based on Vietnamese financial research reports. The benchmark is specifically designed to evaluate the three document representation paradigms proposed in this paper, namely conventional OCR-based retrieval (Method 0), chart-preserving multimodal retrieval (Method 1), and the proposed semantic-enriched chart-aware retrieval framework (Method 2).

A. Data Source

The document corpus is collected from publicly available research reports published by the OCBS Research Center.¹ Unlike annual reports that focus on a single company, OCBS continuously publishes professional financial analyses covering a wide spectrum of the Vietnamese financial market. These reports are regularly updated and contain abundant multimodal information including long textual analyses, financial tables, and various statistical charts.

The collected corpus covers multiple financial categories, including

- Equity research reports;
- Bond market reports;
- Sector analyses;
- Macroeconomic reports;
- Asset allocation reports;
- Investment strategy reports.

Moreover, the reports span numerous industries such as banking, steel, real estate, retail, logistics, technology, manufacturing, energy, and other sectors represented in the Vietnamese stock market.

Since the OCBS repository is continuously updated, the document corpus is designed as a dynamic collection rather than a fixed dataset. For experimental reproducibility, a representative subset of the complete corpus is selected to construct the benchmark used in this study.

B. Benchmark Construction

Instead of evaluating on every available report, we construct a manually curated benchmark that contains representative

financial documents exhibiting diverse layouts, chart styles, and reasoning complexity.

The benchmark is designed to evaluate whether semantic chart understanding improves document retrieval rather than merely improving chart interpretation. Consequently, each evaluation instance is centered around a financial question whose answer depends on both textual information and numerical evidence extracted from charts.

The benchmark currently contains approximately **200 expert-designed evaluation queries** spanning multiple financial domains. Each query is associated with a reference answer and manually annotated evaluation criteria to facilitate quantitative assessment of retrieval quality and reasoning capability, covering multiple reasoning categories.

Table III summarizes the taxonomy of the benchmark.

TABLE III
BENCHMARK QUERY CATEGORIES

Category	Number of Queries
Fact Retrieval	20
Numerical Extraction	20
Trend Analysis	20
Chart Interpretation	100
Cross-document Reasoning	20
Investment Analysis	20

These categories progressively increase the reasoning difficulty from simple factual retrieval to complex financial analysis requiring multiple numerical evidences.

C. Ground Truth Annotation

Each benchmark sample is manually annotated using a structured schema specifically designed for financial RAG evaluation.

Rather than storing only the reference answer, the benchmark decomposes every question into multiple semantic components that enable fine-grained evaluation of both retrieval quality and answer generation.

Each annotation contains the following fields:

- **Document Source:** original report URL.
- **Page ID:** source page containing the evidence.
- **Chart ID:** corresponding chart image identifier.
- **User Query:** evaluation question.
- **Expected Fact:** numerical values that must appear in the answer.
- **Expected Insight:** analytical conclusion inferred from the retrieved evidence.
- **Ground Truth Answer:** reference response combining both facts and insights.
- **Expected Chart JSON:** reference chart specification for visualization.

This decomposition enables independent evaluation of numerical correctness, semantic reasoning, and visualization quality.

¹<https://ocbs.com.vn/trung-tam-phan-tich/>

D. Annotation Workflow

The benchmark construction consists of the following stages:

- 1) Financial reports are collected from the OCBS Research Center.
- 2) Every PDF page is processed using EasyOCR to recover textual information.
- 3) YOLOv8 detects chart regions and extracts chart images.
- 4) Financial experts manually verify the extracted charts and identify the numerical evidence required for downstream evaluation.
- 5) Evaluation questions are constructed according to predefined reasoning categories.
- 6) Ground-truth answers and reference chart specifications are manually created.

This workflow ensures that every evaluation instance is fully traceable to the original financial report.

E. Evaluation Protocol

Unlike traditional document QA benchmarks that evaluate only textual similarity, our benchmark evaluates multiple aspects of a financial RAG system.

The generated answer is compared against the annotated ground truth from four complementary perspectives.

1) *Retrieval Evaluation*: Retrieval quality is measured using standard Information Retrieval metrics including

- Recall@k
- Mean Reciprocal Rank (MRR)
- nDCG

These metrics evaluate whether the correct document pages and chart contexts are successfully retrieved before answer generation.

2) *Answer Evaluation*: The generated textual response is evaluated using the RAGAS framework, including

- Faithfulness;
- Answer Relevance;
- Context Precision;
- Context Recall.

These metrics quantify how faithfully the generated answer is grounded in the retrieved evidence.

3) *Chart Understanding Evaluation*: Since the proposed framework explicitly models chart semantics, additional chart-oriented metrics are introduced.

- **Fact Fidelity**
Measures whether all expected numerical values are correctly reflected in the generated answer.
- **Insight Relevance**
Measures whether the generated financial interpretation agrees with the annotated analytical insight.
- **Data Mapping Score**
Measures the consistency between the generated chart specification and the reference chart representation.
- **Chart Structural Accuracy**

Measures whether the generated visualization correctly preserves chart type, data series, axis semantics, and legend information.

These metrics enable a comprehensive evaluation of the complete financial reasoning pipeline instead of evaluating text generation alone.

F. Discussion

The benchmark is intentionally designed to evaluate document representation rather than chart recognition accuracy alone.

Method 0 is expected to perform well on purely textual questions but suffers from missing visual evidence because charts are discarded during OCR.

Method 1 preserves chart images, allowing multimodal reasoning during inference. However, since chart semantics are not incorporated into document embeddings, retrieval remains primarily dependent on textual similarity.

The proposed Method 2 transforms charts into structured semantic knowledge before indexing. By enriching document representations with numerical facts and analytical insights extracted from charts, the retrieval stage receives substantially richer contextual information, thereby increasing the probability of retrieving relevant evidence for complex financial reasoning tasks. The experiments presented in the following sections are designed to validate this hypothesis quantitatively.

““ x

V. EXPERIMENTAL SETUP

This section presents the experimental configuration used to evaluate the proposed framework. Since the primary objective of this work is to investigate the impact of different document representation strategies on Retrieval-Augmented Generation (RAG), all competing methods are evaluated under the same document corpus, query benchmark, embedding model, and large language model whenever applicable. This ensures that the observed performance differences originate from the document indexing strategy rather than unrelated implementation choices.

A. Compared Methods

Three document representation strategies are evaluated throughout this work.

- **Method 0 (Conventional OCR-based RAG).**

Financial reports are processed using EasyOCR to extract textual content only. Charts are discarded during preprocessing, resulting in a purely text-based document representation. The extracted text is divided into page-level chunks, embedded using a Hugging Face embedding model, and indexed in the Milvus vector database.

- **Method 1 (Chart-Preserving Multimodal RAG).**

In addition to OCR, YOLOv8 is employed to detect chart regions from each document page. Detected chart images are preserved and stored separately inside the retrieval database. During inference, both textual chunks and chart

images are retrieved and provided to the multimodal language model for reasoning.

- **Method 2 (Proposed Semantic-Enriched RAG).**

The proposed framework extends Method 1 by replacing raw chart images with structured semantic representations generated by the chart understanding module. Instead of retrieving chart images directly, numerical values, chart metadata, summaries, and analytical insights extracted from DePlot are fused into the textual representation before indexing. Consequently, the vector database stores semantically enriched document embeddings that provide richer contextual information for dense retrieval.

B. Implementation Details

The proposed framework is implemented in Python using PyTorch and Hugging Face Transformers. All document embeddings are stored in Milvus, while Attu is adopted as the visualization and database management interface.

The major software components are summarized below.

TABLE IV
IMPLEMENTATION CONFIGURATION

Component	Configuration
OCR Engine	EasyOCR
Object Detection	YOLOv8
Chart Understanding	DePlot
Embedding Model	HuggingFace (v0.19.x)
Vector Database	Milvus v2.x
Database GUI	Attu
Generation Model	Gemini (3.5)
Programming Language	Python 3.x
Framework	PyTorch

All experiments are conducted using identical software configurations unless otherwise specified.

C. Hardware Configuration

Experiments are conducted on a workstation equipped with

- CPU: Intel Core i7-13700H
- GPU: NVIDIA GeForce RTX 4050 (Ada Lovelace, 6GB VRAM GDDR6)
- RAM: 16GB GB
- CUDA Version: 12.3
- Operating System: Window 11

To evaluate the deployment cost of the document preprocessing pipeline, GPU memory consumption, execution time, and throughput are reported for each preprocessing stage.

D. OCR Performance Evaluation

Since OCR serves as the first stage of all three methods, its computational cost is analyzed independently.

The evaluation considers

- Processing time per page;
- GPU memory consumption;
- Throughput under different batch sizes;
- Average processing latency.

The OCR benchmark is designed to determine an appropriate batch size that balances processing speed and GPU memory usage during large-scale document indexing.

E. Evaluation Metrics

Following Section IV, retrieval quality is evaluated using Recall@k, Mean Reciprocal Rank (MRR), and nDCG.

Answer quality is evaluated using the RAGAS framework, including Faithfulness, Answer Relevance, Context Precision, and Context Recall.

For chart-aware reasoning, additional metrics including Fact Fidelity, Insight Relevance, Data Mapping Score, and Chart Structural Accuracy are reported.

Furthermore, system efficiency is evaluated by measuring

- Document indexing time;
- Retrieval latency;
- End-to-end response time;
- GPU memory consumption.

F. Research Hypotheses

The experiments are designed to validate three research hypotheses.

- **H1:** Preserving chart images significantly improves retrieval performance compared with conventional OCR-only RAG.
- **H2:** Transforming charts into structured semantic knowledge before indexing further improves retrieval quality compared with preserving raw chart images.
- **H3:** Offline chart understanding reduces online inference latency while maintaining high factual accuracy and answer faithfulness.

G. Embedding Configuration

Unless otherwise specified, all document chunks are embedded using ...

To investigate the influence of embedding models, additional experiments are conducted using multiple publicly available embedding models from Hugging Face.

The compared embedding models include

- all-MiniLM-L6-v2
- BAAI/bge-small-en-v1.5
- BAAI/bge-large-en-v1.5
- ...

The same retrieval pipeline is used for all embedding models.

The quantitative analyses presented in the next section are organized according to these hypotheses.

H. Results and Analysis

summarize the performance of the three retrieval strategies under the two evaluation criteria.

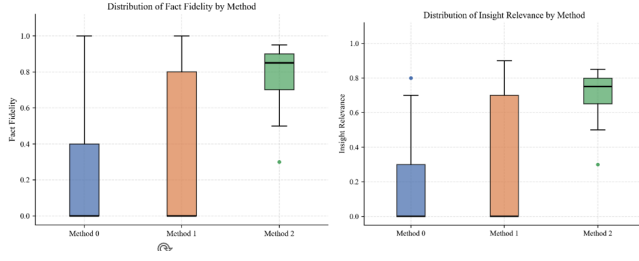


Fig. 6. Average Fact Fidelity and Insight Relevance across three methods. Error bars denote one standard deviation.

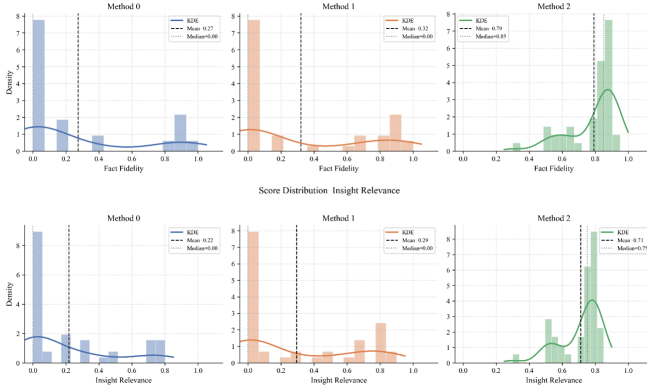


Fig. 7. Score distributions for the three methods. Method 1 exhibits substantially larger variance than Method 2.

1) *Overall Performance*: Method 2 consistently achieves the highest performance on both evaluation metrics. Its average Fact Fidelity reaches 0.791, while Insight Relevance reaches 0.712, substantially outperforming both Method 0 and Method 1.

Compared with the baseline (Method 0), introducing a single retrieved chart (Method 1) produces only modest improvements (Fact Fidelity: 0.273 $\rightarrow$ 0.320, Insight Relevance: 0.218 $\rightarrow$ 0.293). In contrast, the proposed enriched retrieval strategy (Method 2) increases both metrics by approximately 2.5–3 times, indicating that providing richer contextual information enables the language model to produce significantly more accurate factual statements and more relevant analytical insights.

Moreover, statistical hypothesis testing confirms that the improvements of Method 2 over Method 1 are highly significant ($p < 10^{-6}$), accompanied by a large effect size (Cohen’s $d \approx 1.11$), suggesting that the observed gains are not caused by random variation but represent a substantial improvement in practical performance.

TABLE V
OVERALL PERFORMANCE OF THE THREE RETRIEVAL METHODS. HIGHER VALUES INDICATE BETTER PERFORMANCE.

2*Method	Fact Fidelity			Insight Relevance		
	Mean	Median	SD	Mean	Median	SD
Method 0	0.273	0.000	0.379	0.218	0.000	0.287
Method 1	0.320	0.000	0.402	0.293	0.000	0.352
Method 2	0.791	0.850	0.153	0.712	0.750	0.127

TABLE VI
PAIRWISE STATISTICAL SIGNIFICANCE USING THE WILCOXON SIGNED-RANK TEST.

Metric	Comparison	p-value	Cohen’s d	Effect
3*Fact Fidelity	M0 vs M1	0.042	-0.344	Small
	M1 vs M2	$< 10^{-6}$	-1.114	Large
	M0 vs M2	$< 10^{-6}$	-1.288	Large
3*Insight Relevance	M0 vs M1	0.027	-0.375	Small
	M1 vs M2	$< 10^{-6}$	-1.113	Large
	M0 vs M2	$< 10^{-6}$	-1.552	Large

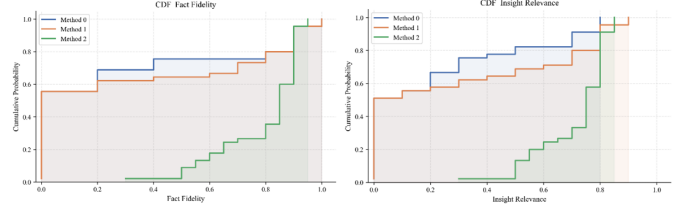


Fig. 8. Score CDF for the three methods. Method 1 exhibits substantially larger variance than Method 2.

2) *Performance Variability*: Although Method 1 achieves slightly higher average scores than the baseline, its performance exhibits considerably larger variance, as illustrated by the boxplots and score distributions in Figures ?? and ?. The standard deviation of Method 1 is substantially higher than that of Method 2 for both Fact Fidelity and Insight Relevance, indicating that its retrieval quality is highly inconsistent across benchmark samples.

This phenomenon can be explained by the retrieval mechanism employed in Method 1. Instead of discarding charts as in Method 0, the system retrieves chart images together with OCR text. Consequently, when the retriever successfully identifies the chart relevant to the user query, the downstream multimodal language model receives sufficient visual evidence and is capable of producing highly accurate answers, resulting in scores close to one. However, if the retrieved chart is only partially related or completely irrelevant to the query, the language model loses access to the required numerical evidence, causing the generated response to deteriorate dramatically. As a result, Method 1 produces a polarized score distribution containing both near-perfect successes and complete failures, which explains its large variance despite only moderate improvements in average performance.

In contrast, Method 2 consistently achieves both higher

TABLE VII
DISTRIBUTION STATISTICS ILLUSTRATING SCORE STABILITY. LOWER CV
INDICATES MORE STABLE PERFORMANCE.

Method	Metric	IQR	Variance	CV
Method 0	Fact Fidelity	0.40	0.143	1.385
Method 1	Fact Fidelity	0.80	0.161	1.255
Method 2	Fact Fidelity	0.20	0.023	0.194
Method 0	Insight Relevance	0.30	0.082	1.318
Method 1	Insight Relevance	0.70	0.124	1.200
Method 2	Insight Relevance	0.15	0.016	0.179

average scores and substantially lower variance. Most benchmark samples are concentrated within a relatively narrow high-score region, indicating that the proposed retrieval strategy produces not only more accurate but also significantly more stable responses.

3) *Discussion on Retrieval Context Enrichment*: The experimental results suggest that the superiority of Method 2 originates primarily from improvements in the *retrieval stage* rather than the generation stage. Unlike Method 1, which simply retrieves chart images and forwards them to the multimodal language model, Method 2 first transforms each chart into a structured semantic representation consisting of chart meta-data, numerical values, chart type, captions, and automatically generated analytical insights. These chart-derived descriptions are then indexed together with the OCR-extracted document text within a unified retrieval corpus.

This design substantially enriches the semantic context available to the retriever. Financial questions involving trends, comparisons, extrema, temporal changes, or numerical relationships often contain very few lexical overlaps with the original PDF text. Consequently, conventional text retrieval frequently fails to identify the correct evidence. By explicitly converting visual information into searchable textual representations, Method 2 greatly increases the probability that semantically relevant chunks are retrieved for chart-oriented queries.

The richer retrieval context directly benefits the downstream language model. Since the retrieved evidence already contains structured numerical information and high-level chart semantics, the language model no longer needs to infer these relationships solely from raw chart images. Instead, it can focus on reasoning over well-grounded evidence, leading to significant improvements in both Fact Fidelity and Insight Relevance.

This observation is further supported by the pairwise improvement analysis, where Method 2 consistently outperforms Method 1 across the majority of benchmark samples. Rather than improving only a small subset of queries, the enriched retrieval corpus provides systematic gains across diverse chart-related question types, demonstrating that contextual augmentation at the indexing stage is considerably more effective than relying exclusively on multimodal reasoning during inference.

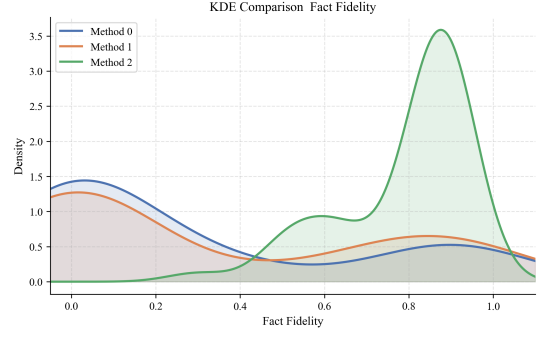


Fig. 9. KDE Fact Fidelity

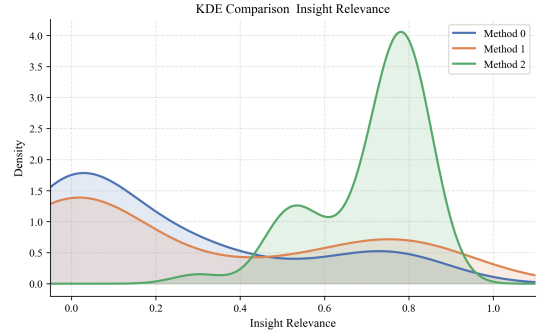


Fig. 10. KDE Insight Relevance

4) *Relationship Between Fact Fidelity and Insight Relevance*: All three methods exhibit strong positive Pearson correlations ($r > 0.89$), indicating that improvements in factual correctness are consistently associated with more relevant analytical insights.

Method 2 achieves the strongest correlation ($r = 0.987$), suggesting that once reliable contextual evidence is provided, the language model is able to simultaneously improve factual accuracy and high-level reasoning. This observation implies that the proposed retrieval framework enhances not only information grounding but also downstream reasoning quality.

5) *Threshold Analysis*: Threshold analysis further highlights the robustness of the proposed method.

For stringent quality thresholds (e.g., scores above 0.8), Method 2 retains a substantially larger proportion of successful samples, whereas the success rates of Method 0 and Method 1 decrease rapidly.

This result indicates that Method 2 not only improves average performance, but also consistently produces high-quality responses across a large portion of the benchmark.

6) *Pairwise Improvement Analysis*: Pairwise score differences provide additional evidence of the effectiveness of the proposed retrieval strategy.

Compared with Method 1, Method 2 produces positive score improvements for the overwhelming majority of evaluation samples, with only a small number of cases showing negligible performance differences.

The improvement histograms demonstrate that the gains are broadly distributed across the dataset rather than being concentrated on a few examples. Therefore, the superior average performance of Method 2 is achieved through consistent improvements over most benchmark instances instead of isolated outliers. Overall, the experimental results consistently demonstrate that enriched chart retrieval substantially improves both factual grounding and analytical reasoning.

While introducing a single retrieved chart can occasionally produce highly accurate responses, its effectiveness strongly depends on retrieval success, leading to considerable performance instability. By incorporating richer contextual evidence, Method 2 not only achieves significantly higher average scores but also greatly reduces score variance, yielding a retrieval framework that is both more accurate and more robust for chart-based question answering.

VI. LIMITATIONS

Although the proposed framework demonstrates substantial improvements over conventional RAG systems, several limitations remain.

First, the current benchmark is constructed exclusively from Vietnamese financial reports. Consequently, the proposed retrieval pipeline has not yet been validated on reports written in other languages or following different reporting standards.

Second, the quality of chart understanding still depends on the performance of the chart extraction model. Extremely low-resolution charts, overlapping graphical elements, or unconventional visual styles may reduce extraction accuracy and subsequently affect downstream retrieval.

Third, the current implementation employs page-level chunking for document indexing. While this strategy simplifies the indexing pipeline, finer-grained chunking strategies or hierarchical retrieval mechanisms may further improve retrieval precision.

Finally, although Method 2 significantly enriches retrieval context, the additional preprocessing introduces higher offline indexing costs. Future work should investigate more efficient indexing strategies while maintaining retrieval quality.

VII. FUTURE WORK

Several promising research directions remain open.

Future work will investigate graph-enhanced retrieval by integrating knowledge graphs into the retrieval stage, enabling more effective cross-document reasoning. We also plan to explore stronger multilingual embedding models specifically optimized for Vietnamese financial terminology. Another promising direction is adaptive chunking based on document layout instead of fixed page-level segmentation. Finally, incorporating more advanced chart understanding models and expanding the benchmark with additional financial document types will further improve the robustness and generalization capability of the proposed framework.

VIII. CONCLUSION

This paper presented a comparative study of three Retrieval-Augmented Generation pipelines for Vietnamese financial document understanding. Starting from a conventional OCR-based RAG system, we progressively incorporated multimodal chart retrieval and chart-aware semantic indexing to investigate the impact of visual information on financial question answering.

Experimental results demonstrate that merely providing chart images to the language model improves performance only marginally and leads to highly unstable retrieval behavior. In contrast, transforming chart contents into structured semantic representations before indexing substantially enriches the retrieval corpus, allowing the retriever to identify more relevant evidence for chart-oriented queries. Consequently, the proposed Method 2 consistently achieves the highest Fact Fidelity and Insight Relevance while significantly reducing score variance.

These findings highlight that the primary benefit of chart understanding lies in improving the retrieval stage rather than increasing the reasoning capability of the language model itself. We believe that integrating structured visual semantics into Retrieval-Augmented Generation represents a promising direction for multimodal financial document understanding and can be extended to other document-intensive domains such as healthcare, legal analysis, and scientific literature.

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